

TECHBIZ SECURITY

CYBER SECURITY INTERNSHIP REPORT

Week 1 – Task 1

Cyber Security Intro + What is a SOC?

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1. Introduction

This report has been prepared as part of Week 1, Task 1 of my SOC Analyst internship at TechBiz Security. The purpose of this task was to build a foundational understanding of core cybersecurity concepts before moving into hands-on Security Operations Center (SOC) work. As a fresher entering the cybersecurity field, it was important for me to first understand why organisations invest so heavily in protecting their information systems, and what role a SOC plays in that effort.

Cybersecurity today is not limited to installing antivirus software or setting up a firewall. It is a continuous process of identifying risks, protecting assets, detecting malicious activity, and responding to incidents before they cause serious damage. Almost every organisation, regardless of size, depends on digital infrastructure, which means every organisation is also a potential target for cyber attacks. Understanding the fundamentals of cybersecurity is therefore the first and most essential step for anyone starting a career in this field.

2. Objectives

The objectives of this task were as follows:

- To understand the basic principles of cybersecurity and why they matter.
- To learn the CIA Triad (Confidentiality, Integrity, Availability) and how it forms the foundation of security decision-making.
- To understand the difference between a threat, a vulnerability, and a risk.
- To learn what a Security Operations Center (SOC) is and what functions it performs.
- To understand the different SOC roles — L1, L2, L3 analysts, Incident Responders, and Threat Hunters.
- To understand the difference between Blue Team and Red Team functions in an organisation.

3. Cybersecurity Fundamentals

Cybersecurity is the practice of protecting systems, networks, applications, and data from unauthorized access, damage, or disruption. It combines technology, processes, and people to defend an organisation's digital assets against attackers who may try to steal information, disrupt operations, or cause financial and reputational harm.

During this task, I studied the CIA Triad, which is the central model used across the cybersecurity industry to guide security decisions. The three pillars of the CIA Triad are:

- **Confidentiality** — Ensuring that information is only accessible to authorised individuals. This is achieved through mechanisms such as encryption, access control, and authentication.
- **Integrity** — Ensuring that data is accurate and has not been tampered with, either accidentally or maliciously. Hashing and checksums are common ways to verify integrity.
- **Availability** — Ensuring that systems and data are accessible to authorised users whenever needed. This involves redundancy, backups, and protection against denial-of-service attacks.

Why the CIA Triad is Important

The CIA Triad is important because it gives security teams a simple, structured way to evaluate any security control or incident. For example, when a new system is being designed, security teams ask: does this protect confidentiality, does it preserve integrity, and does it maintain availability? Almost every security control that exists — from firewalls to backups to access permissions — maps back to one or more of these three principles.

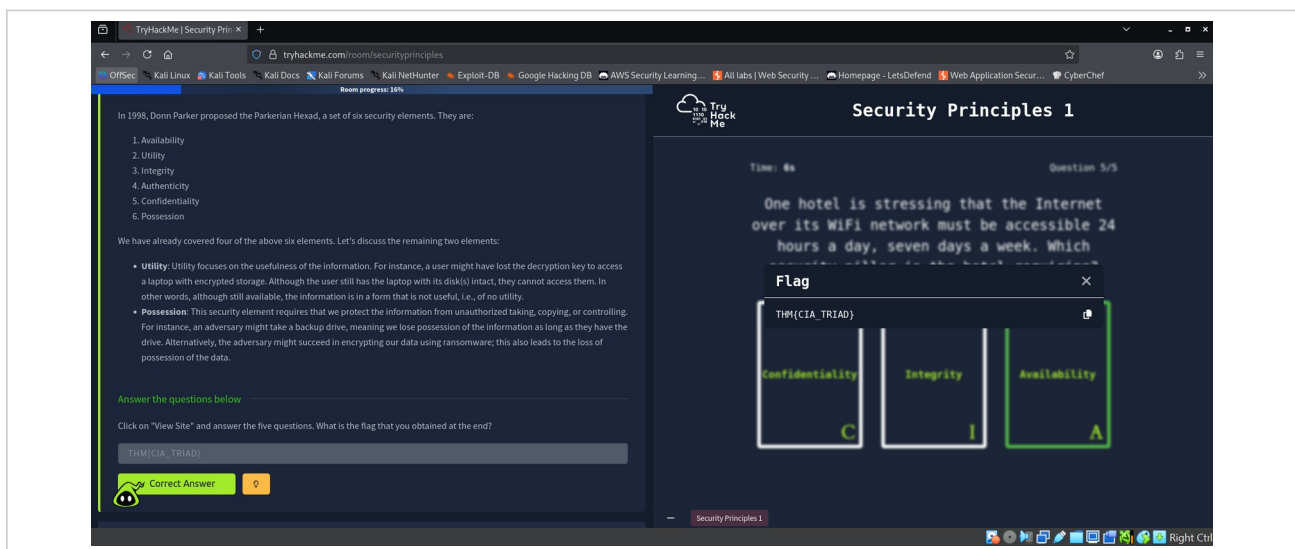


Figure 1: TryHackMe Security Principles – CIA Triad Learning Evidence

This screenshot shows the completion of the CIA Triad exercise on the TryHackMe Security Principles room, including the correctly obtained flag THM{CIA_TRIAD}. This confirms my practical understanding of Confidentiality, Integrity, and Availability as applied to a real-world scenario.

4. Threat, Vulnerability and Risk

One of the most important distinctions I learned during this task was the difference between a threat, a vulnerability, and a risk. These three terms are often used interchangeably by beginners, but they mean very different things in a professional security context.

- **Threat** — Any potential event or actor that could cause harm to a system, such as a hacker, malware, or a natural disaster.
- **Vulnerability** — A weakness in a system, application, or process that could be exploited by a threat, such as an unpatched software flaw or a weak password policy.
- **Risk** — The potential for loss or damage when a threat actually exploits a vulnerability. Risk is generally understood as a combination of likelihood and impact.

A simple real-world example: if an organisation is running an outdated web server (vulnerability), and an attacker on the internet is actively scanning for such servers (threat), then the risk is the possibility that the organisation's server gets compromised, potentially leading to data loss or service disruption.

Aspect	Threat	Vulnerability	Risk
Definition	A potential danger or malicious actor	A weakness that can be exploited	The possible impact/loss if exploited
Example	A hacker or malware	Unpatched software	Data breach leading to financial loss
Controllable?	Mostly external, hard to control	Can be fixed/patched	Can be reduced/managed

5. Security Operations Center (SOC)

A Security Operations Center (SOC) is a centralised team within an organisation that is responsible for continuously monitoring, detecting, analysing, and responding to cybersecurity threats. The SOC acts as the first line of defence, working around the clock to make sure that suspicious activity is identified and dealt with before it turns into a major security incident.

Purpose of a SOC

The primary purpose of a SOC is to reduce the time between when an attack begins and when it is detected and stopped. The longer an attacker remains undetected inside a network, the more damage they can cause. A well-functioning SOC significantly reduces this window of exposure.

Main Responsibilities of a SOC

- Security monitoring — Continuously watching network traffic, endpoints, and logs for unusual activity.
- Log analysis — Reviewing logs generated by firewalls, servers, and applications to identify patterns of compromise.
- Alert investigation — Reviewing alerts raised by security tools to determine if they represent real threats or false positives.
- Incident detection — Identifying that a security incident is occurring or has occurred.
- Incident response — Taking action to contain, eradicate, and recover from a security incident.
- Threat intelligence — Gathering and using information about known attackers, malware, and attack techniques.
- Security reporting — Documenting findings and communicating them to relevant stakeholders.

SOC Workflow

From what I studied, the general SOC workflow follows a consistent cycle:

Monitor → Detect → Analyse → Investigate → Respond → Document

It is important to note that this task focused on learning the theory and structure of SOC operations. I have not personally performed live SOC monitoring or handled real incidents at this stage — this understanding was built through structured learning material and guided practical exercises.

6. SOC Roles and Responsibilities

A SOC is typically organised into tiers, with each level handling increasingly complex tasks. During this task I learned about the following roles:

Role	Main Responsibility	Typical Activities	Escalation Level
SOC Analyst L1	First-line monitoring and triage	Reviewing alerts, basic log checks, escalating suspicious events	Escalates to L2
SOC Analyst L2	In-depth investigation	Correlating logs, analysing malware behaviour, confirming incidents	Escalates to L3
SOC Analyst L3	Advanced analysis & mentoring	Deep forensic analysis, tuning detection rules, guiding L1/L2	Escalates to IR team
Incident Responder	Managing active incidents	Containment, eradication, recovery, coordinating response efforts	Reports to management
Threat Hunter	Proactive threat discovery	Searching for hidden threats not caught by automated alerts	Feeds findings to SOC

Each of these roles requires a growing set of skills — L1 analysts typically need strong fundamentals in networking and log analysis, while L3 analysts and threat hunters require deeper knowledge of malware analysis, scripting, and attacker techniques.

7. Blue Team vs Red Team

Another important concept covered in this task was the distinction between Blue Team and Red Team functions in cybersecurity.

- **Blue Team** — The defensive side of security. Their objective is to protect the organisation by monitoring systems, detecting attacks, and strengthening defences. SOC teams are part of the Blue Team.
- **Red Team** — The offensive side of security. Their objective is to simulate real attacks against the organisation to identify weaknesses before actual attackers do. Penetration testers are part of the Red Team.

Aspect	Blue Team	Red Team
Objective	Defend systems and detect attacks	Simulate attacks to find weaknesses
Approach	Defensive security	Offensive security
Typical Activities	Monitoring, log analysis, incident response	Penetration testing, exploitation, social engineering
Goal	Minimise damage and downtime	Expose vulnerabilities before real attackers do

Both teams work towards the same overall goal — a more secure organisation — but from opposite directions. The Red Team finds the gaps, and the Blue Team closes them. Many organisations also run a 'Purple Team' function where Red and Blue teams collaborate directly to improve detection and response capability.

To build a supporting understanding of the offensive/Red Team side of security, I also completed the TryHackMe Pentesting Fundamentals room, which introduced the basic stages and ethics of penetration testing.

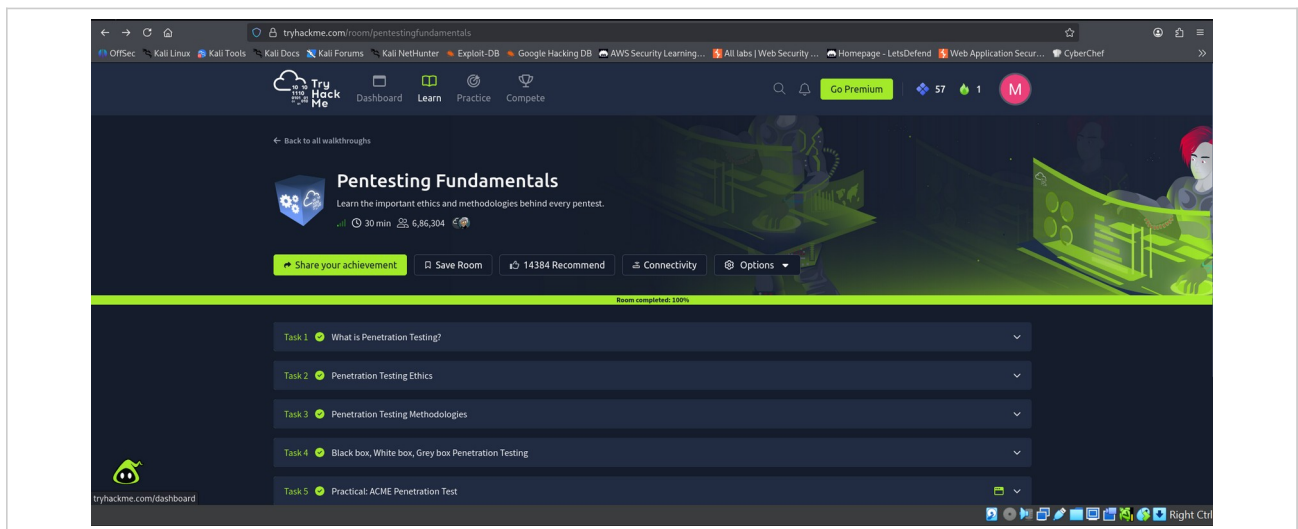


Figure 2: TryHackMe Penetration Testing Fundamentals – Learning Evidence

This screenshot shows the Pentesting Fundamentals room at 100% completion, covering what penetration testing is, testing ethics, methodologies, and black/white/grey box testing — supporting my understanding of Red Team (offensive security) concepts.

8. Practical Learning Evidence

This section presents the screenshots collected as evidence of the practical learning activities completed during this task.

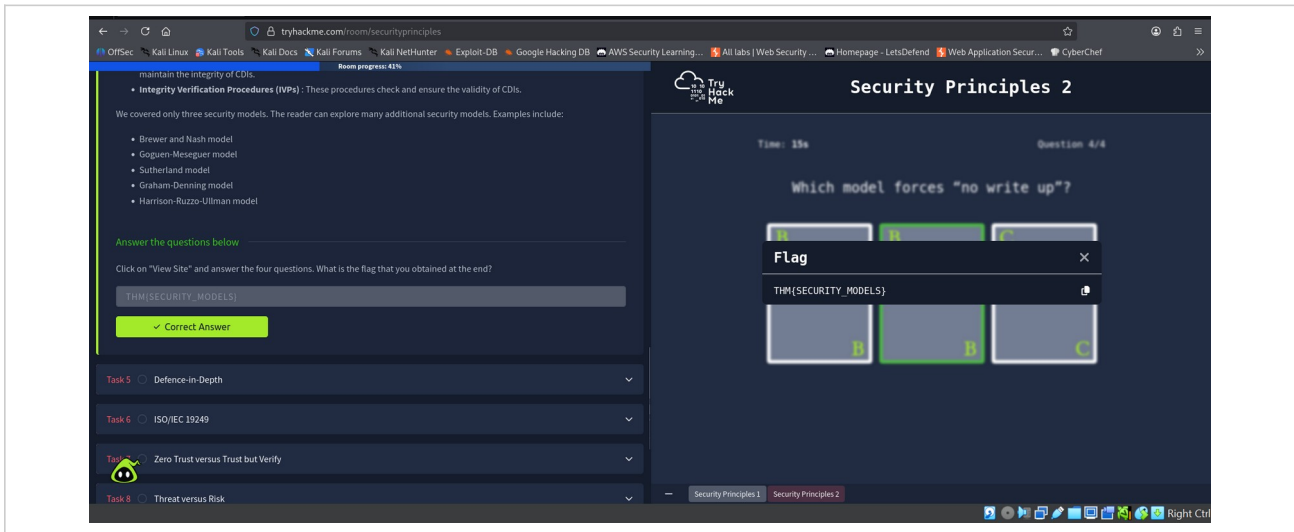


Figure 3: TryHackMe Security Principles – Security Models Exercise

This screenshot shows completion of the Security Models exercise within the Security Principles room, with the flag THM{SECURITY_MODELS} obtained after answering questions on concepts such as the Brewer and Nash, Sutherland, and Graham-Denning models.

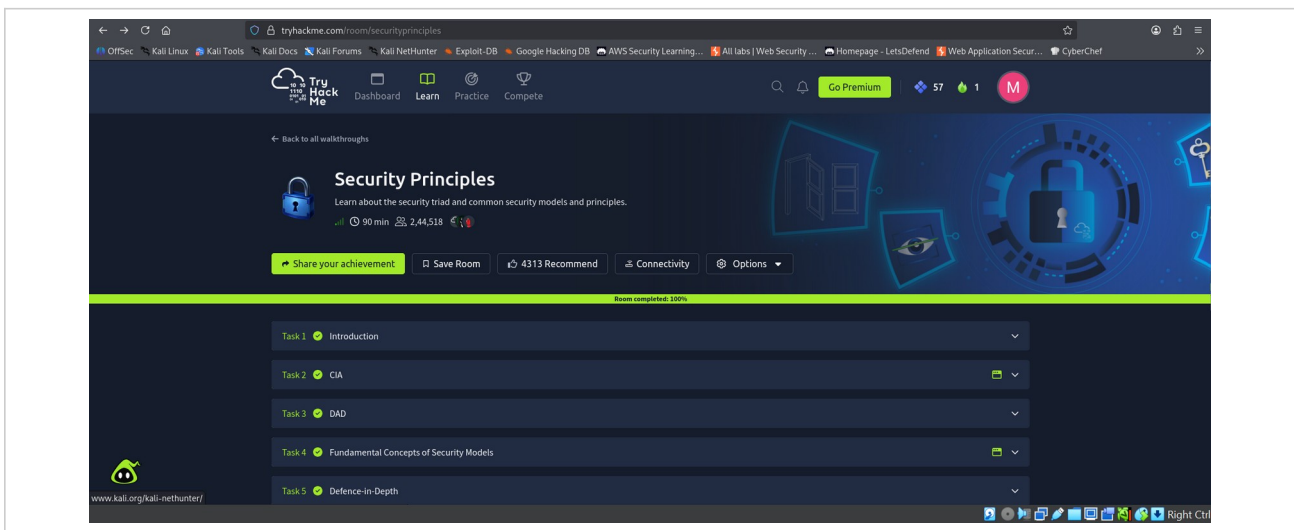


Figure 4: TryHackMe Security Principles – Room 100% Completed

This screenshot confirms that the Security Principles room was completed in full, including the tasks on Introduction, CIA, DAD, and Fundamental Concepts of Security Models.

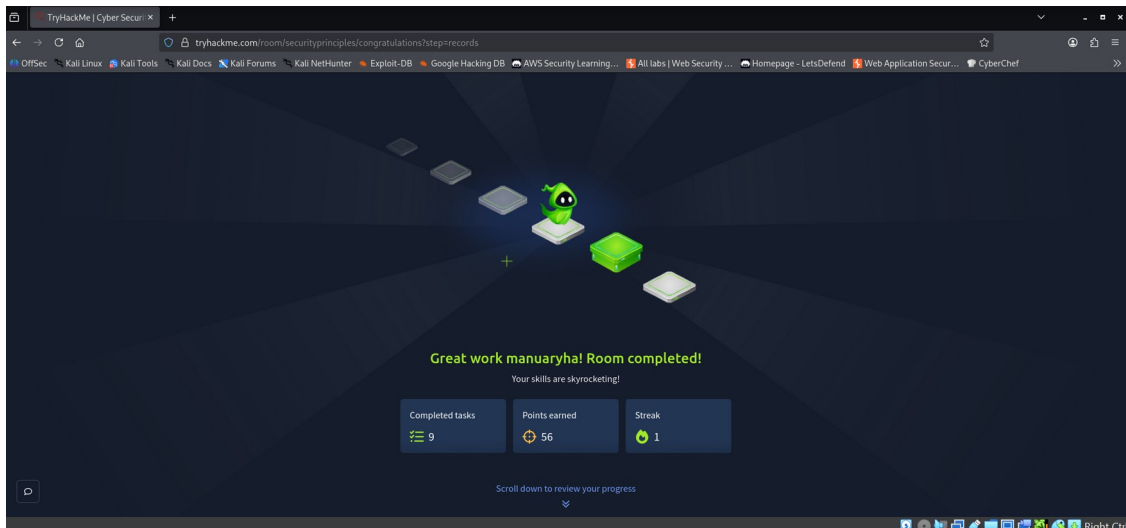


Figure 5: TryHackMe Security Principles – Completion Summary

This screenshot shows the final completion summary for the Security Principles room, with 9 tasks completed and 56 points earned.

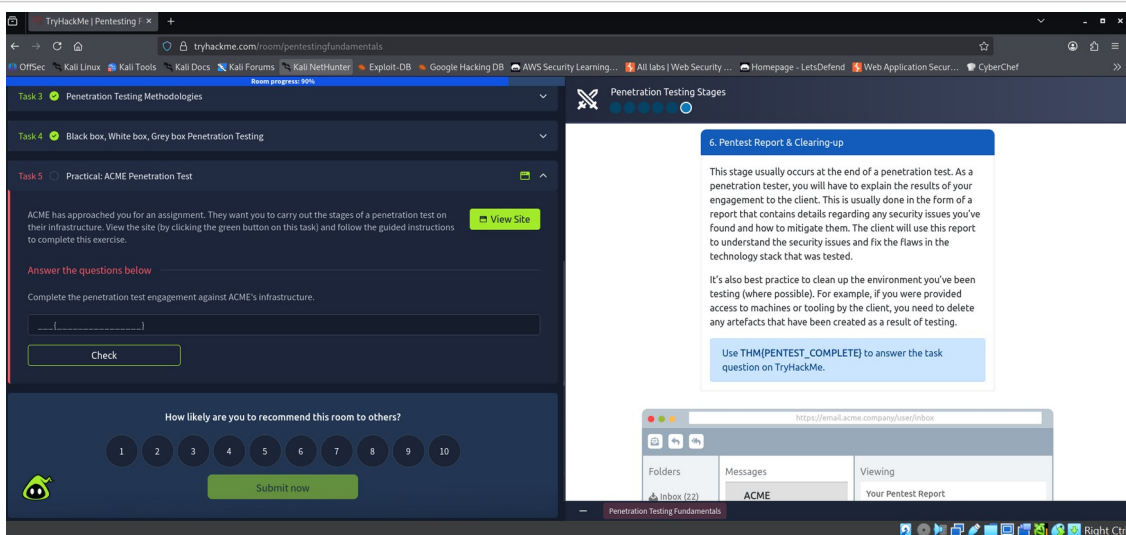


Figure 6: TryHackMe Pentesting Fundamentals – Practical ACME Engagement

This screenshot shows the practical exercise where ACME's infrastructure was used to walk through the stages of a penetration test, concluding with the Pentest Report and Clean-up stage and the flag THM{PENTEST_COMPLETE}.

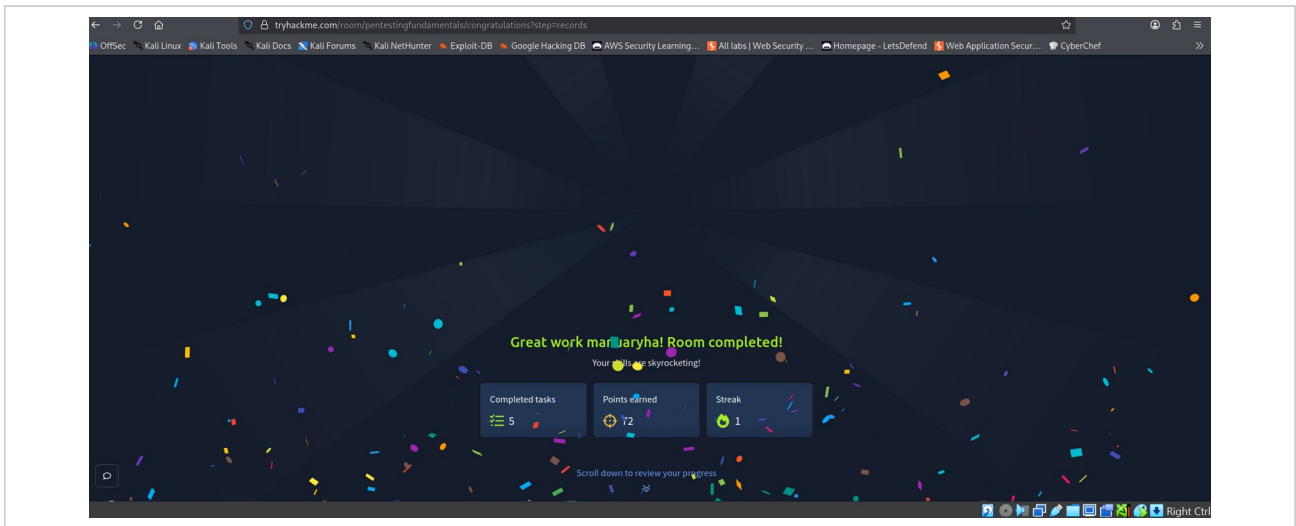


Figure 7: TryHackMe Pentesting Fundamentals – Completion Summary

This screenshot shows the final completion summary for the Pentesting Fundamentals room, with 5 tasks completed and 72 points earned.

9. Key Learnings

Through this task, I understood the following major concepts:

- The fundamentals of cybersecurity and why it matters for modern organisations.
- The CIA Triad — Confidentiality, Integrity, and Availability — as the foundation of security thinking.
- The clear distinction between a threat, a vulnerability, and a risk.
- The purpose, responsibilities, and workflow of a Security Operations Center (SOC).
- The different SOC roles, from L1 analyst to Threat Hunter, and how escalation works between them.
- The difference between Blue Team (defensive) and Red Team (offensive) security functions.
- The basic stages and ethics involved in penetration testing, as a supporting concept to Red Team operations.

10. Conclusion

This first task gave me a strong conceptual foundation for the rest of my SOC Analyst internship. I learned that cybersecurity is not just about tools, but about a structured way of thinking — evaluating confidentiality, integrity, and availability, understanding the difference between threats, vulnerabilities and risks, and knowing how a SOC team operates to detect and respond to incidents. From this exercise, I understood how SOC roles are structured in tiers and how Blue Team and Red Team functions complement each other to keep an organisation secure. I feel well prepared to build on these fundamentals in the upcoming tasks of my internship.

11. References

- TryHackMe — Security Principles (tryhackme.com/room/securityprinciples)
- TryHackMe — Penetration Testing Fundamentals (tryhackme.com/room/pentestingfundamentals)